

Interactive-Shiny-report-of-HTSanalyzeR2

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Package

HTSanalyzeR2 0.99.19

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1 Introduction

In this short tutorial, we will give a detailed illustration for the interactive Shiny report of **HTSanalyzeR2** to visualize the results and modify figures in different aspects.

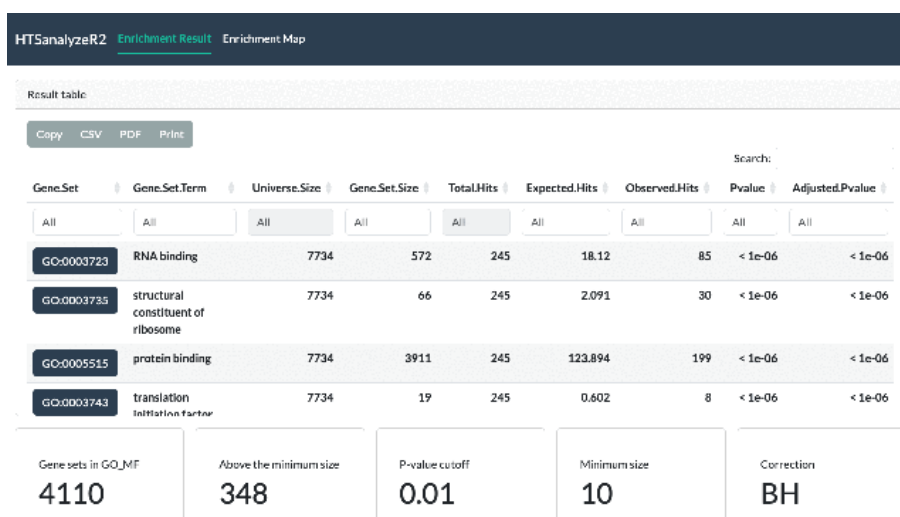
2 Interactive Shiny report visualization

2.1 Visualize single GSCA object for individual data set

For single data set analysis, after performing gene set analysis by HTSanalyzeR2, we can get a GSCA object and use the function *report* to launch the Shiny report.

```
data(d7_gsca)
report(d7_gsca)
```

2.1.1 The GSOA result table [Figure1]



The screenshot shows the HTSanalyzeR2 Enrichment Result table. It includes a search bar and a table with columns: Gene.Set, Gene.Set.Term, Universe.Size, Gene.Set.Size, Total.Hits, Expected.Hits, Observed.Hits, Pvalue, and Adjusted.Pvalue. The table lists four gene sets: GO:0003723 (RNA binding), GO:0003735 (structural constituent of ribosome), GO:0005515 (protein binding), and GO:0003743 (translation initiation factor). Below the table, there are summary statistics: Gene sets in GO_MF (4110), Above the minimum size (348), P-value cutoff (0.01), Minimum size (10), and Correction (BH).

Gene.Set	Gene.Set.Term	Universe.Size	Gene.Set.Size	Total.Hits	Expected.Hits	Observed.Hits	Pvalue	Adjusted.Pvalue
GO:0003723	RNA binding	7734	572	245	18.12	85	< 1e-06	< 1e-06
GO:0003735	structural constituent of ribosome	7734	66	245	2.091	30	< 1e-06	< 1e-06
GO:0005515	protein binding	7734	3911	245	123.894	199	< 1e-06	< 1e-06
GO:0003743	translation initiation factor	7734	19	245	0.602	8	< 1e-06	< 1e-06

Gene sets in GO_MF: 4110
 Above the minimum size: 348
 P-value cutoff: 0.01
 Minimum size: 10
 Correction: BH

Figure 1: The GSOA result table

The above screenshot shows the GSOA result table where users can either get a general summary of the analysis parameters, choose which analysis and which gene set collection to show or download the result by different format.

2.1.2 Parameters for modifying the figure [Figure2]

The settings live in a collapsible panel beside every graph. It has five sections: **Label**, **Nodes**, **Edges**, **Colour scheme** and **Layout**. Every change is applied to the rendered graph in place, without recomputing the analysis or resetting the layout.

- **Label:** how nodes are labelled, with three parameters: *Text* (gene set ID, term, or none), *Label scale* and *Label colour*.
- **Nodes:** *Scale*, *Opacity*, *Border width*, *Border colour* and the colour used for nodes without a phenotype value.
- **Edges:** *Scale* and *Colour*.

- **Colour scheme:** the two ends of the positive and of the negative colour scale. The sign of the enrichment score or phenotype decides which scale a node uses.
- **Layout:** the node-repulsion parameters of the force-directed layout.
 - Mode: *Lin-Log mode* and *Strong gravity mode* change how the layout treats distance.
 - Parameters: *Edge repel* and *Adjust sizes* make the figure looser.
 - Gravity: ranging from -50 to 50. The larger it is, the looser the pattern.

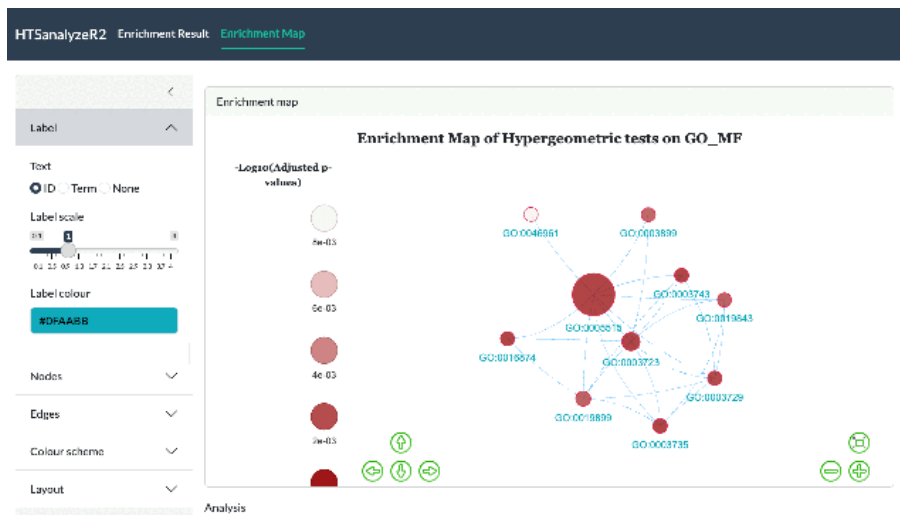


Figure 2: The enrichment map of GSOA result

The above screenshot shows the enrichment map of a GSOA result together with the settings panel. Hovering a node shows its identifier, term and value; the buttons in the lower right pan, zoom, fit the view and export the graph as PNG.

2.2 Visualize single *NWA* object for individual data set

For single data set analysis, after performing enriched subnetwork analysis by HTSanalyzeR2, we can get a *NWA* object and use the function *report* to launch the Shiny report.

```
data(d7_nwa)
report(d7_nwa)
```

2.2.1 The identified subnetwork [Figure3]

The below screenshot shows the identified subnetwork where users can either get a general summary of the subnetwork attributes, modify the subnetwork based on their data and preference or download the subnetwork as PNG for further use.

2.3 Visualize a list of *GSCA* objects for “Time-course” data

```
data(gscats)
## To make the figure more compact, we set a cutoff
## to move any other edges with low Jaccard coefficient.
reportAll(gscats, cutoff = 0.03)
```

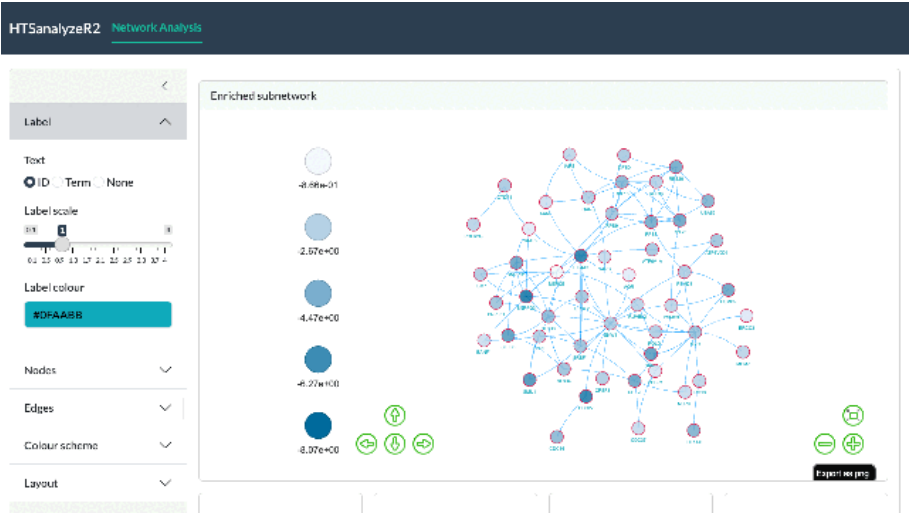


Figure 3: The identified subnetwork

2.3.1 GSEA results table for “Time-course” data [Figure4]

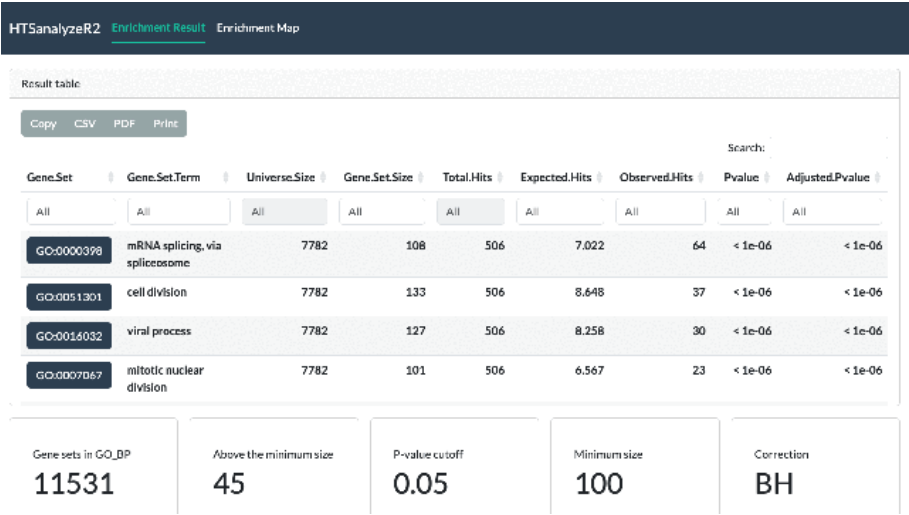


Figure 4: GSEA results table for “Time-course” data

The above screenshot shows the GSEA result tables for Time-course data with three time points where users can either get a general summary of the analysis parameters, choose which analysis, which gene set collection and which experiment result to show or download the result by different format.

2.3.2 Union enrichment map for “Time-course” data [Figure5]

The below screenshot shows the union enrichment maps of GSEA result for “Time-course” data with filtered edges by setting a cutoff on the edges. The *Experiment* slider moves through the time points while the layout is kept fixed, so the three maps can be compared directly. Here, we can clearly see a gradual change among them.

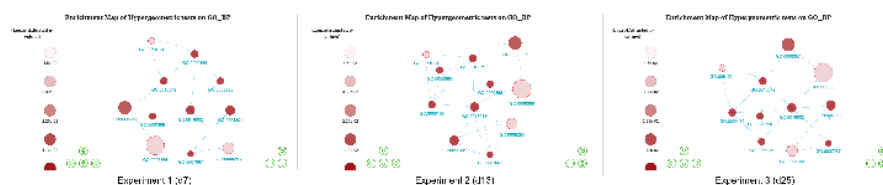


Figure 5: Union enrichment map for “Time-course” data

2.4 Visualize a list of *NWA* objects for “Time-course” data

```
data(nwaTS)
reportAll(nwa = nwaTS)
```

2.4.1 Union subnetwork for “Time-course” data [Figure6]

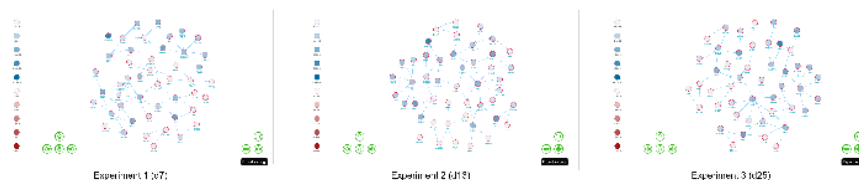


Figure 6: Union subnetwork for “Time-course” data

2.5 Visualize both *GSCA* and *NWA* objects simultaneously

```
reportAll(gsca = gscaTS, nwa = nwaTS)
```

The above screenshot shows the union subnetworks for “Time-course” data. As with the enrichment maps, the *Experiment* slider moves through the time points while the layout stays fixed, so a gradual change among the three union subnetworks can be seen clearly.

2.5.1 Visualize both *GSEA* and network result in the same report [Figure7]

If you do both gene set analysis and enriched subnetwork analysis, it’s also possible to visualize both of them in the same Shiny report as showed here.

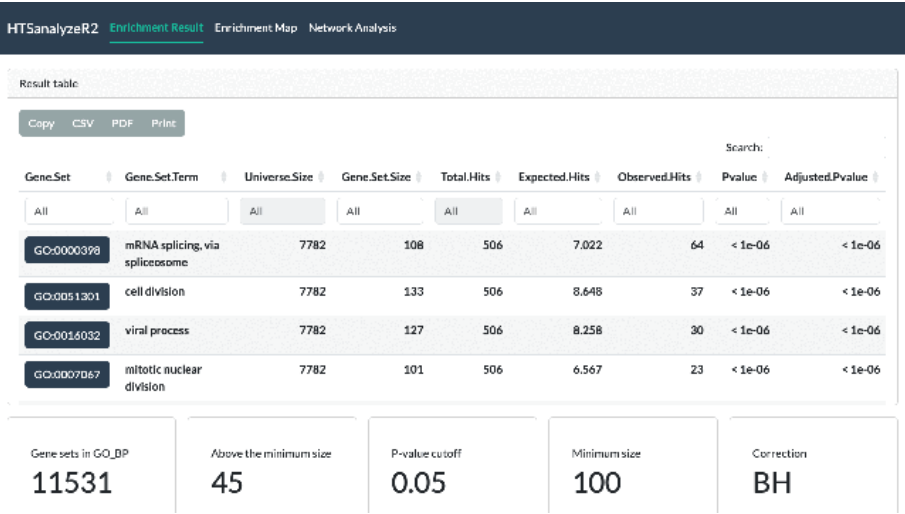


Figure 7: Visualize both GSEA and network result in the same report

3 Session Info

```
## R version 4.6.0 (2026-04-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.12
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8
##
## time zone: Asia/Shanghai
## tzcode source: internal
##
## attached base packages:
## [1] stats4 stats graphics grDevices utils datasets methods
## [8] base
##
## other attached packages:
## [1] TxDb.Hsapiens.UCSC.hg19.knownGene_3.22.1
## [2] GenomicFeatures_1.64.0
## [3] GenomicRanges_1.64.0
## [4] Seqinfo_1.2.0
## [5] limma_3.68.5
## [6] igraph_2.3.3
## [7] GO.db_3.23.1
## [8] KEGGREST_1.52.2
## [9] org.Hs.eg.db_3.23.1
## [10] AnnotationDbi_1.74.0
## [11] IRanges_2.46.0
```

```
## [12] S4Vectors_0.50.0
## [13] Biobase_2.72.0
## [14] BiocGenerics_0.58.0
## [15] generics_0.1.4
## [16] HTSanalyzeR2_0.99.19
## [17] BiocStyle_2.40.0
##
## loaded via a namespace (and not attached):
## [1] DBI_1.3.0                bitops_1.1-0
## [3] rlang_1.3.0             magrittr_2.0.5
## [5] shinydashboard_0.7.3    msigdbr_26.1.1
## [7] otel_0.2.0              matrixStats_1.5.0
## [9] compiler_4.6.0          RSQLite_3.53.3
## [11] png_0.1-9               vctrs_0.7.3
## [13] stringr_1.6.0           pkgconfig_2.0.3
## [15] crayon_1.5.3            fastmap_1.2.0
## [17] XVector_0.52.0          Rsamtools_2.28.0
## [19] promises_1.5.0          rmarkdown_2.31
## [21] tinytex_0.59            bit_4.6.0
## [23] xfun_0.60               cachem_1.1.0
## [25] Rmpfr_1.1-2             cigarillo_1.2.1
## [27] blob_1.3.0              later_1.4.8
## [29] DelayedArray_0.38.2     gmp_0.7-5.1
## [31] BiocParallel_1.46.0     parallel_4.6.0
## [33] R6_2.6.1                stringi_1.8.7
## [35] RColorBrewer_1.1-3      rtracklayer_1.72.0
## [37] Rcpp_1.1.1-1.1          bookdown_0.48
## [39] assertthat_0.2.1        SummarizedExperiment_1.42.0
## [41] iterators_1.0.14        knitr_1.51
## [43] RankProd_3.38.0         httpuv_1.6.17
## [45] Matrix_1.7-5            tidyselect_1.2.1
## [47] abind_1.4-8             yaml_2.3.12
## [49] codetools_0.2-20        miniUI_0.1.2
## [51] curl_7.1.0              lattice_0.22-9
## [53] tibble_3.3.1            shiny_1.14.0
## [55] S7_0.2.2                evaluate_1.0.5
## [57] Biostrings_2.80.2       pillar_1.11.1
## [59] BiocManager_1.30.27     MatrixGenerics_1.24.0
## [61] DT_0.34.0               foreach_1.5.2
## [63] RCurl_1.98-1.20         ggplot2_4.0.3
## [65] scales_1.4.0            BioNet_1.72.0
## [67] xtable_1.8-8            glue_1.8.1
## [69] tools_4.6.0             BiocIO_1.22.0
## [71] data.table_1.18.2.1     fgsea_1.38.0
## [73] colourpicker_1.3.0      GenomicAlignments_1.48.0
## [75] XML_3.99-0.24           fastmatch_1.1-8
## [77] cowplot_1.2.0           grid_4.6.0
## [79] restfulr_0.0.17         cli_3.6.6
## [81] S4Arrays_1.12.0         dplyr_1.2.1
## [83] gtable_0.3.6            digest_0.6.39
## [85] SparseArray_1.12.2      rjson_0.2.23
```

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```
## [87] htmlwidgets_1.6.4      farver_2.1.2
## [89] memoise_2.0.1          htmltools_0.5.9
## [91] lifecycle_1.0.5        http_1.4.9
## [93] statmod_1.5.2          mime_0.13
## [95] bit64_4.8.0
```